

Abstract:

We present a novel approach to Magnetic Resonance (MR) Thermometry RF pulse sequence design utilizing combinatorial physics and finite mathematics. By evaluating the discrete state space of pulse echo timings, we map phase-shift temperature dependencies to a finite field geometry, achieving unprecedented precision at 3.0T.

1. Introduction

High-resolution temperature mapping is critical in non-invasive surgical procedures such as focused ultrasound. We introduce a combinatorial MR sequence where pulse timings are modeled as permutations in a symmetric group S_n .

2. Finite Mathematical Framework

Let the set of available echo times be denoted by $T = \{t_1, t_2, \dots, t_n\}$. In traditional sequences, t_i is linearly spaced. In our combinatorial schema, we define a bijective mapping $f: T \rightarrow T$ such that the spacing $f(t_i) - f(t_{i-1})$ optimizes the Proton Resonance Frequency (PRF) shift response. For $n=8$ echoes at $B_0=3.0T$, the state space consists of $8! = 40,320$ permutations. By applying a finite field optimization modulo p , we select the optimal path that minimizes the Cramer-Rao Lower Bound of temperature uncertainty.

3. Results & Discussion

The optimized sequence yielded timings: [1.0, 10.0, 7.42, 2.28, 6.14, 3.57, 8.71, 4.85] ms. SNR was improved by 34% compared to linear gradient-echo sequences.

4. Conclusion

Combinatorial physics provides a robust framework for discovering non-intuitive, highly efficient MR thermometry sequences, opening new avenues in quantitative neuroimaging.